

MACHINE LEARNING PROJECT

Machine Learning

Credit risk classification: from preprocessing to SAFE AI metrics

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Project workflow

OVERVIEW

01

Preprocessing

Feature engineering,
encoding and scaling

02

Unsupervised

Clustering and PCA
visualisation

03

Supervised

Classification and
model comparison

04

SAFE AI

RGE and RGR
robustness metrics

Data preparation

Clean the raw Lending Club variables and create financially meaningful predictors.

Model evaluation

Compare predictive performance and investigate the robustness of the best-performing model.

Feature engineering

PREPROCESSING

Transform raw variables

New predictors were created to capture leverage, credit history and borrower quality.

Examples

Credit age

Credit usage ratio

Loan-to-income ratio

Average FICO

```
df["credit_age"] = 2024 - df["earliest_cr_line"].dt.year

df["credit_usage_ratio"] = (
    df["revol_bal"] / (df["total_rev_hi_lim"] + 1)
)

df["loan_to_income_ratio"] = (
    df["loan_amnt"] / (df["annual_inc"] + 1)
)

df["fico_avg"] = (
    df["fico_range_low"] + df["fico_range_high"]
) / 2
```

Selected features

PREPROCESSING

01

Loan characteristics

Loan amount
Interest rate
Installment
Grade
DTI

02

Credit profile

FICO score
Credit age
Open accounts
Total accounts
Public records

03

Engineered features

Credit usage
Record indicator
Recent activity
Loan / income
Average FICO

Ordinal variables

ENCODING

Label encoding

Loan grade is ordinal, so its natural ranking should be retained.

$$A > B > C > D > E > F > G$$

The categories are mapped to a decreasing numerical scale.

```
df_filtered["grade"] = (  
    df_filtered["grade"]  
    .str.strip()  
    .str.upper()  
)  
  
mapping_grade = {  
    "A": 6, "B": 5, "C": 4, "D": 3,  
    "E": 2, "F": 1, "G": 0  
}  
  
df_filtered["grade"] = (  
    df_filtered["grade"].map(mapping_grade)  
)
```

Nominal variables

ENCODING

One-hot encoding

Home ownership is nominal, therefore no ranking should be imposed.

Mortgage	Rent
Own	Any
Other	None

```
df_filtered = pd.get_dummies(  
    df_filtered,  
    columns=["home_ownership"],  
    drop_first=False  
)  
  
dummy_cols = [  
    "home_ownership_MORTGAGE",  
    "home_ownership_RENT",  
    "home_ownership_OWN",  
    "home_ownership_ANY",  
    "home_ownership_OTHER",  
    "home_ownership_NONE"  
)  
  
df_filtered[dummy_cols] = df_filtered[dummy_cols].astype(int)
```

Missing values

DATA QUALITY

```
print(df_filtered.info())  
  
print("Null values in dataset:")  
  
print(  
    df_filtered  
    .isnull()  
    .sum()  
)
```

Missing observations were inspected before the modelling stage.

Missing values detected

Variable	Missing
DTI	0.028%
inq_last_6mths	< 0.001%
credit usage ratio	5.02%
public record score	0.052%
recent activity score	5.02%

Largest concentration: approximately 5%.

Exploratory data analysis

EDA

Distribution analysis

- Histograms
- Boxplots
- Frequency plots
- Outlier detection

Dependence analysis

- Correlation matrix
- Heat map
- Feature relationships
- Target behaviour

EDA was used to understand distributions, outliers and relationships among predictors.

Standardisation or normalisation?

SCALING

Standardisation

$$z_i = \frac{x_i - \mu}{\sigma}$$

Centers the variable around zero and rescales it to unit standard deviation.

Normalisation

$$x_i^* = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}}$$

Maps the variable into a fixed interval, typically $[0, 1]$.

Our choice: Standardisation

Target variable

CLASS IMBALANCE

Fully paid



Charged off



5:1

Fully-paid loans are approximately five times more frequent than charged-off loans.

Class imbalance must therefore be considered when evaluating classification performance.

Clustering

UNSUPERVISED LEARNING

Cluster summary

```
cluster_summary = (  
    X_sample  
    .groupby("cluster")  
    .mean(numeric_only=True)  
)  
  
print(cluster_summary)
```

Group averages help interpret the characteristics of each cluster.

PCA visualisation

```
pca = PCA(n_components=2)  
  
X_sample_pca = pca.fit_transform(X_sample)  
  
pca_df = pd.DataFrame(  
    X_sample_pca,  
    columns=["PC1", "PC2"]  
)  
  
pca_df["cluster"] = km.labels_
```

PCA reduces the feature space to two dimensions for visualisation.

Regularisation

SUPERVISED LEARNING

Ridge — L2

$$\mathcal{L}_{Ridge} = \mathcal{L}_0 + \lambda \sum_{j=1}^p \beta_j^2$$

Shrinks large coefficients towards zero.

Lasso — L1

$$\mathcal{L}_{Lasso} = \mathcal{L}_0 + \lambda \sum_{j=1}^p |\beta_j|$$

Can reduce some coefficients exactly to zero.

Logistic Regression

MODEL RESULTS

	Predicted 0	Predicted 1
True 0	103	63
True 1	43	111

66.9%

Accuracy

Precision	0.638
Recall	0.721
F1-score	0.677
AUC	0.73

Neural Network

MODEL RESULTS

	Predicted 0	Predicted 1
True 0	115	51
True 1	33	121

73.8%

Accuracy

Precision	0.703
Recall	0.786
F1-score	0.742
AUC	0.80

K-Nearest Neighbours

MODEL RESULTS

	Predicted 0	Predicted 1
True 0	105	61
True 1	13	141

76.9%

Accuracy

Precision	0.698
Recall	0.916
F1-score	0.792
AUC	0.88

Highest recall

Decision Tree

MODEL RESULTS

	Predicted 0	Predicted 1
True 0	127	39
True 1	31	123

78.1%

Accuracy

Precision	0.759
Recall	0.799
F1-score	0.778
AUC	0.78

Random Forest

MODEL RESULTS

	Predicted 0	Predicted 1
True 0	147	19
True 1	18	136

88.4%

Accuracy

Precision	0.877
Recall	0.883
F1-score	0.880
AUC	0.94

Best overall model

XGBoost

MODEL RESULTS

	Predicted 0	Predicted 1
True 0	142	24
True 1	24	130

85.0%

Accuracy

Precision	0.844
Recall	0.844
F1-score	0.844
AUC	0.93

Second-best AUC

Support Vector Classifier

MODEL RESULTS

	Predicted 0	Predicted 1
True 0	109	57
True 1	42	112

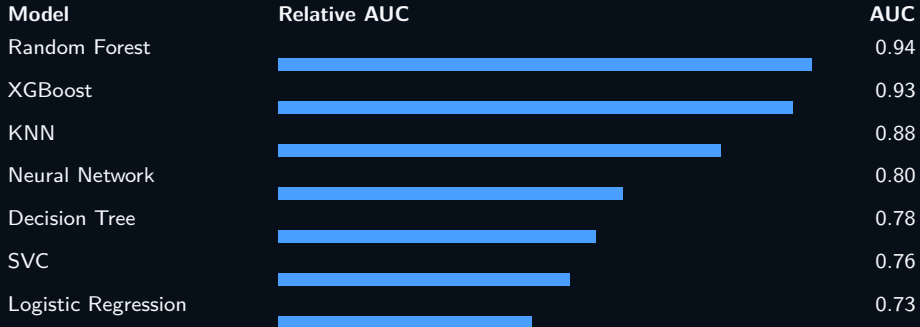
69.1%

Accuracy

Precision	0.663
Recall	0.727
F1-score	0.693
AUC	0.76

Model performance

AUC COMPARISON



Model comparison

SUMMARY

Model	Accuracy	Precision	Recall	F1-score
Random Forest	0.884	0.877	0.883	0.880
XGBoost	0.850	0.844	0.844	0.844
KNN	0.769	0.698	0.916	0.792
Decision Tree	0.781	0.759	0.799	0.778
Neural Network	0.738	0.703	0.786	0.742
SVC	0.691	0.663	0.727	0.693
Logistic Regression	0.669	0.638	0.721	0.677

Random Forest provides the strongest overall performance, while **KNN** achieves the highest recall.

RGE — Random Forest

SAFE AI

Highest-ranked features

1. Grade
2. Inquiries last 6 months
3. Loan-to-income ratio
4. Interest rate
5. Open accounts
6. Annual income
7. FICO range high
8. DTI
9. Credit age
10. Average FICO

Lower-ranked features

11. Loan amount
12. Installment
13. FICO range low
14. Total accounts
15. Delinquencies
16. Home ownership
17. State
18. Public records
19. Record indicator

RGR — Random Forest

SAFE AI

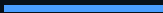
Feature	RGR
home_ownership_ANY	1.000000
has_record	0.999822
addr_state	0.999820
home_ownership_RENT	0.999779
pub_rec	0.999621
home_ownership_MORTGAGE	0.999531
home_ownership_OWEN	0.999526
public_record_score	0.999375
dti	0.998518
delinq_2yrs	0.998287
open_acc	0.998001
total_acc	0.997780

Feature	RGR
credit_age	0.996909
fico_range_low	0.996448
loan_to_income_ratio	0.996413
installment	0.996086
fico_avg	0.996059
inq_last_6mths	0.994902
annual_inc	0.994707
loan_amnt	0.994420
fico_range_high	0.994172
int_rate	0.990059
grade	0.983514

MACHINE LEARNING PROJECT

Thank you

for your attention



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